GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Independent Set 6.07 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

No — 6/4 = 15/10 = 1.5, but 20/12 ≈ 1.67

Two ratios agree and the third does not.

2. A triangle has sides 8, 10 and 13. A similar triangle's 10-side is 25. Find its other two sides.

20 and 32.5

k = 25 ÷ 10 = 2.5.

3. A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

3 and 5

k = 6 ÷ 30 = 0.2.

4. A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

10.5 m

k = 14 ÷ 2 = 7; 1.5 × 7.

5. Two similar right triangles share corner A. The small one runs 3 m along the ground and stands 1.2 m high. The big one's upright is 9 m beyond the small one's. How high is it?

4.8 m

Its base is 3 + 9 = 12 m from A: k = 4.

6. Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

12, 15, 21 — every side × 1.5

10, 12, 16 adds 2 to every side.